Exploring heterogeneity in meta-analysis: Meta-regression analysis

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Meta-regression is simply the extension of the regression analysis. Whereas in the regression analysis, the outcome and the covariates are measured for each subject, in meta-regression, the outcome and the covariates are measured on the level of the study. Because studies differ at least in the sample size, we need to assign a weight in each study under a specific model (fixed-effect or random-effects model) before implementing a meta-regression analysis. Weight assignment is not addressed in a regression analysis, because we study subjects. The differences between the regression and the meta-regression analysis are obvious, but the goal is the same: to assess the association between an outcome and one or more covariates.

AN INTRODUCTION TO META-REGRESSION

Meta-regression (and regression) is a versatile tool. In a meta-regression model, we can examine a set of covariates as well as their transformations. A meta-regression model with more than 1 covariates is known as multiple meta-regression model. We have the flexibility to examine simultaneously different types of covariates and assess whether there is an association between them. A meta-regression model with 1 categorical covariate yields the same results with the subgroup analysis. However, in practice, we rarely work with more than 1 covariate in the meta-regression model owing to limitations in the number of the included studies. We need a fairly large ratio of studies to covariates to yield reliable results. Therefore, the more covariates we want to assess, the more studies we need to include in the meta-analysis. We use a real example to illustrate the meta-regression analysis.

APPLICATION TO REAL DATA–TOTAL SAMPLE SIZE

In this worked example, we will examine whether there is an association between the effect size and the total sample size. It has been shown that studies with smaller sample sizes tend to provide larger effect sizes than studies with larger sample sizes; this "behavior" is known as a small-study effect, and we will discuss it in the upcoming article in this column.

We will use again the binary data of the 5 studies[1-5] included in a meta-analysis of a published systematic review that compared bond failures between plasma and halogen curing lights.[6] Because of the small number of studies, it is impossible to evaluate more than 1 covariate. The summary data of the studies are presented in Table I. The measure of the effect of interest is the risk ratio. First, we will analyze the effect sizes in the natural logarithm scale and report the results in the risk ratio scale after exponentiation. The risk ratio ranges from 0.71 (reduction in risk by 29%) to 1.75 (increase in risk by 75%), and this implies possibly substantial variation in the effect sizes across the studies. Some of this variation is due to the inherent sampling error in each study, and some of it might be attributed to real variations in the true effects. We will attempt to explain part of this variation by assessing the association between the effect size and the total sample size of the studies in a random-effects meta-regression model.

Table II presents the results of the meta-regression model. The regression coefficient is equal to 0.9997 (measured in the risk ratio scale), and it indicates that the total sample size of the studies has a negligible impact on the effect sizes; as the total sample size increases, the effect size decreases negligibly by 0.03% (ie, $(0.9997 - 1) \times 100 = -0.03\%$). Essentially, the regression coefficient quantifies the magnitude of the association between the dependent variable Y (here, the effect size measures as risk ratio) and the covariate X (here, the total sample size): regression coefficient

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Table I. Bond failures between plasma and halogen curing lights

	Plasma			*Halogen*			
Study, year	*Events*	*Nonevents*	*Sample size*	*Events*	*Nonevents*	*Sample size*	*Total sample*
Cacciafesta et al, 2004[1]	21	279	300	12	288	300	600
Manzo et al, 2004[2]	12	292	304	12	292	304	608
Pettemerides et al, 2004[3]	12	164	176	13	163	176	352
Russell et al, 2008[4]	22	305	327	31	296	327	654
Sfondrini et al, 2004[5]	31	686	717	39	678	717	1434

Table II. Results from the meta-regression analysis of risk ratio against total sample size

Component	*Value*
Regression coefficient	0.9997
95% confidence interval	0.998−1.001
z-test (*P* value)	−0.75 (0.45)

equal to 0 implies no association, positive regression coefficient implies a positive association between the variables (as X increases, so does Y), and negative regression coefficient implies negative association between the variables (as X increases, Y decreases).

In our example, the 95% confidence interval ranges from 0.998 to 1.001, and it includes the value 1, which implies no statistical association between total sample size and effect size. Furthermore, the (2-sided) z-test tests the null hypothesis that the regression coefficient is equal to 0, against the alternative that the regression coefficient is different from 0. The z-test statistic is equal to −0.75 with P value 0.45, and thus, we conclude that the impact of the sample size on the effect sizes is not statistically significant.

BUBBLE PLOT

Bubble plot is a simple plot that illustrates the size and scatter of the trials alongside the prediction line. Figure 1 illustrates a bubble plot. Each study is represented by a *green* bubble, with the sizes of the bubbles proportional to the weight assigned to the studies: the larger the weight, the larger the bubble. When we use a random-effects model, we expect the weights to be balanced across the studies. The location of each bubble corresponds to the coordinates of the respective study. The slope of the line reflects the direction of the association between the effect sizes and the publication year; as years go by, the effectiveness of the compared interventions tends to decrease. The predicted effect size of each study is projected vertically on this line (indicated by the *black* circles). This line is known as a prediction line. The closer the *green* bubbles to the prediction line, the smaller the difference between the observed and predicted estimates of the included studies, and the better the fit of the meta-regression model to the data.

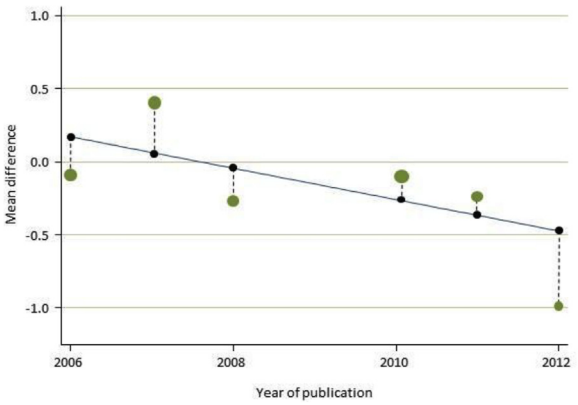


Fig 1. A bubble plot that illustrates the scatter of the mean difference against the year of publication.

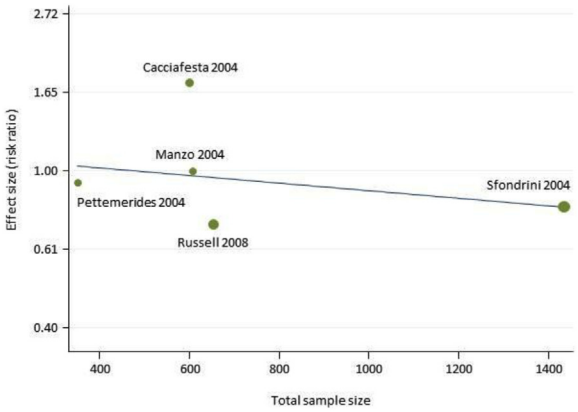


Fig 2. A bubble plot that illustrates the scatter of the effect size (measured as risk ratio) against the total sample size.

In our example, the bubble plot in Figure 2 illustrates the negative but negligible association between the total sample size and the effect sizes in the risk ratio scale. Note that we do not extrapolate our conclusions to sample sizes outside from the range of the data. If we include a trial with total sample size outside the range of the

aforementioned data, we need to perform the meta-regression analysis once again while including the new trial.

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